

58. Data Operation Circuit (DOC)

This is the DOC_B version of the DOC peripheral module.

DOC_B is referred to as DOC in this chapter.

58.1 Overview

The data operation circuit (DOC) is used to compare, add, and subtract 16 or 32-bit data. An interrupt can be generated when the following conditions apply.

- When the 16 or 32-bit compared values match the detection condition
- When the result of 16 or 32-bit data addition overflows
- When the result of 16 or 32-bit data subtraction underflows

Table 58.1 lists the data operation circuit specifications and Figure 58.1 shows a block diagram of the data operation circuit.

Table 58.1 DOC specifications

Item	Description
Data operation function	• 16 or 32-bit data comparison, comparison to detect data above or below thresholds, and window comparison • 16 or 32-bit data addition, and subtraction
Module-stop function	The module-stop state can be set to reduce power consumption.
Interrupts	• The compared values match the detection condition • The result of data addition is greater than 0xFFFF (DOCR.DOBW = 0) or 0xFFFF_FFFF (DOCR.DOBW = 1) • The result of data subtraction is less than 0x0000 (DOCR.DOBW = 0) or 0x0000_0000 (DOCR.DOBW = 1)
Event link function (output)	• The result of data comparison is consistent with detection condition • The result of data addition is greater than 0xFFFF (DOCR.DOBW = 0) or 0xFFFF_FFFF (DOCR.DOBW = 1) • The result of data subtraction is less than 0x0000 (DOCR.DOBW = 0) or 0x0000_0000 (DOCR.DOBW = 1)
TrustZone Filter	Security and Privilege attribution can be set

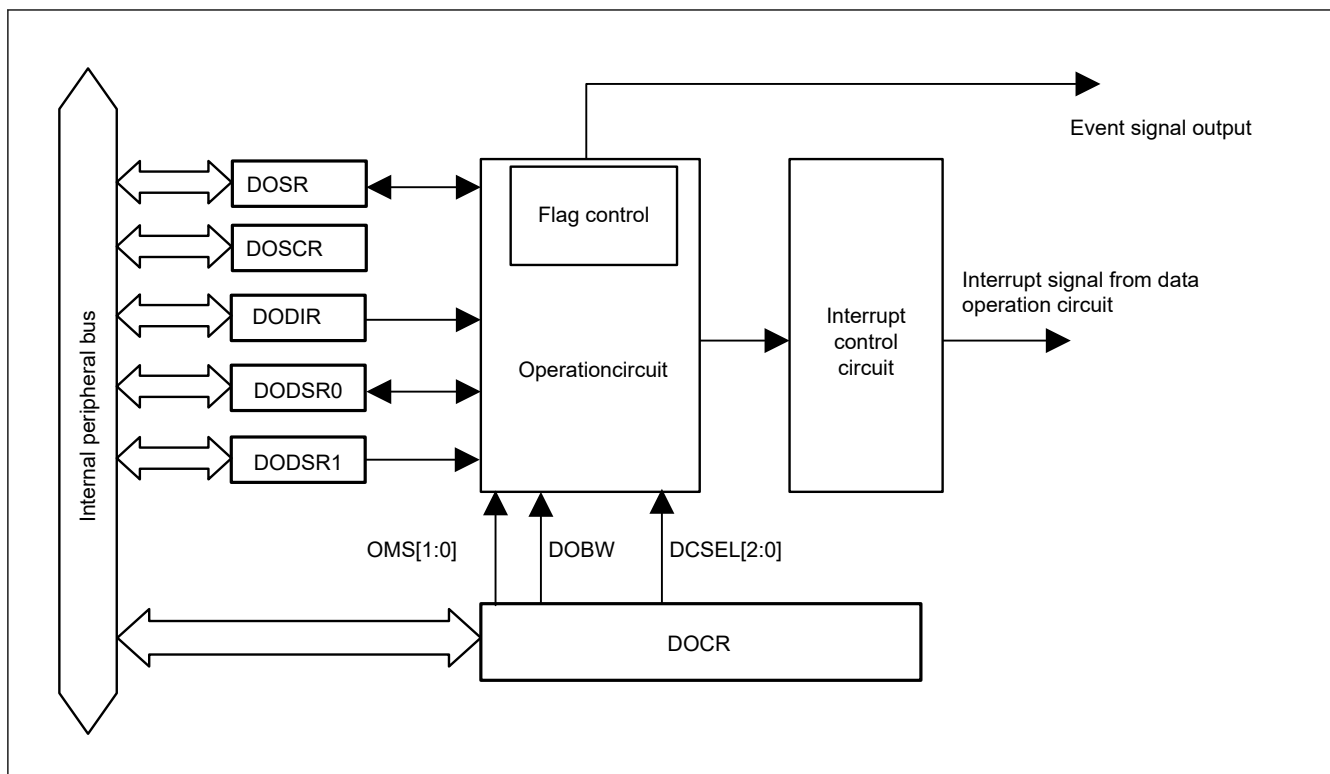


Figure 58.1 DOC block diagram

58.2 DOC Register Descriptions

58.2.1 DOCR : DOC Control Register

Base address: DOC_B = 0x4031_1000
DOC_B_NS = 0x5031_1000

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	DCSEL[2:0]			DOBW	—	OMS[1:0]	

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	OMS[1:0]	Operating Mode Select 0 0: Data comparison mode 0 1: Data addition mode 1 0: Data subtraction mode 1 1: Setting prohibited	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	DOBW	Data Operation Bit Width Select 0: 16-bit 1: 32-bit	R/W
6:4	DCSEL[2:0] ^{*1}	Detection Condition Select 0 0 0: Mismatch (DODSR0 ≠ DODIR) 0 0 1: Match (DODSR0 = DODIR) 0 1 0: Lower (DODSR0 > DODIR) 0 1 1: Upper (DODSR0 < DODIR) 1 0 0: Inside window (DODSR0 < DODIR < DODSR1) 1 0 1: Outside window (DODIR < DODSR0, DODSR1 < DODIR) Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. Valid only when data comparison mode is selected.

The DOCR is a register which can set the operation mode of data operation circuit and interrupt enable/disable.

OMS[1:0] bits (Operating Mode Select)

These bits select the operating mode of the data operation circuit.

DOBW bit (Data Operation Bit Width Select)

This bit selects the bit width of data operation.

DCSEL[2:0] bits (Detection Condition Select)

These bits are valid only when data comparison mode is selected.

These bits select the condition for detection in data comparison mode.

58.2.2 DOSR : DOC Flag Status Register

Base address: DOC_B = 0x4031_1000
DOC_B_NS = 0x5031_1000

Offset address: 0x04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DOPCF

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DOPCF	Data Operation Circuit Flag Indicates the result of an operation.	R
7:1	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

The DOSR register indicates the status of the data operation.

DOPCF flag (Data Operation Circuit Flag)

[Setting conditions]

- DOCR.OMS[1:0] bits = 00b (Data comparison mode): The compared value matches the detection condition selected by DOCR.DCSEL[2:0] bits
- DOCR.OMS[1:0] bits = 01b (Data addition mode): The result of data addition is greater than 0xFFFF (DOCR.DOBW = 0) or 0xFFFF_FFFF (DOCR.DOBW = 1)
- DOCR.OMS[1:0] bits = 10b (Data subtraction mode): The result of data subtraction is less than 0x0000 (DOCR.DOBW = 0) or 0x0000_0000 (DOCR.DOBW = 1)

[Clearing condition]

- Writing 1 to the DOSCR.DOPCFCL bit

58.2.3 DOSCR : DOC Flag Status Clear Register

Base address: DOC_B = 0x4031_1000
DOC_B_NS = 0x5031_1000

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DOPC FCL

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	DOPCFCL	DOPCF Clear 0: Maintains the DOPCF flag state. 1: Clears the DOPCF flag.	W
7:1	—	The write value should be 0.	W

Note: S-TYPE3, P-TYPE3

The DOSCR is a register which can clear the status of data operation. This register is read as 0x00.

DOPCFCL bit (DOPCF Clear)

Setting this bit to 1 clears the DOPCF flag.

58.2.4 DODIR : DOC Data Input Register

Base address: DOC_B = 0x4031_1000
DOC_B_NS = 0x5031_1000

Offset address: 0x0C

Bit position: 31 0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	It stores data used in the operations. Access the DODIR with the bit width of data operation selected by the DOCR.DOBW bit.	R/W

Note: S-TYPE3, P-TYPE3

58.2.5 DODSR0 : DOC Data Setting Register 0

Base address: DOC_B = 0x4031_1000
DOC_B_NS = 0x5031_1000

Offset address: 0x10

Bit position: 31 0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Access the DODSR0 with the bit width of data operation selected by the DOCR.DOBW bit. This register stores data for use as a reference in data comparison mode. When selecting window comparison (DOCR.DCSEL[2:0] = 100b, 101b), set a value less than DODSR1 (DODSR1 > DODSR0). This register also stores the results of operations in data addition and data subtraction modes.	R/W

Note: S-TYPE3, P-TYPE3

58.2.6 DODSR1 : DOC Data Setting Register 1

Base address: DOC_B = 0x4031_1000
DOC_B_NS = 0x5031_1000

Offset address: 0x14

Bit position: 31

0

Bit field:

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Access the DODSR1 with the bit width of data operation selected by the DOCR.DOBW bit. This register stores data for use as a reference in data comparison mode. When selecting window comparison (DOCR.DCSEL[2:0] = 100b, 101b), set a value greater than DODSR0 (DODSR1 > DODSR0). This register is only used for window comparisons.	R/W

Note: S-TYPE3, P-TYPE3

58.3 Operation

58.3.1 Data Comparison Mode

Figure 58.2 to Figure 58.7 shows an example of the steps involved in data comparison mode operation by the data operation circuit.

The following is an example of operation when the bit width of data operation is 32-bit.

1. Writing 00b to the DOCR.OMS[1:0] bits selects data comparison mode, and setting the DOCR.DCSEL[2:0] to selects detection condition.
2. The 32-bit reference data is set in DODSR0 and DODSR1.*1
3. 32-bit data for comparison is written to DODIR.
4. If a value written to DODIR match the detection condition set by DOCR.DCSEL[2:0], the DOCR.DOPCF flag is set to 1 and an ELC event and a data operation circuit interrupt are generated.

Note: The comparison operation is executed only by writing to the DODIR

Note 1. The DODSR1 register setting is required only when window comparison is selected. Set a value greater than DODSR0 (DODSR1 > DODSR0).

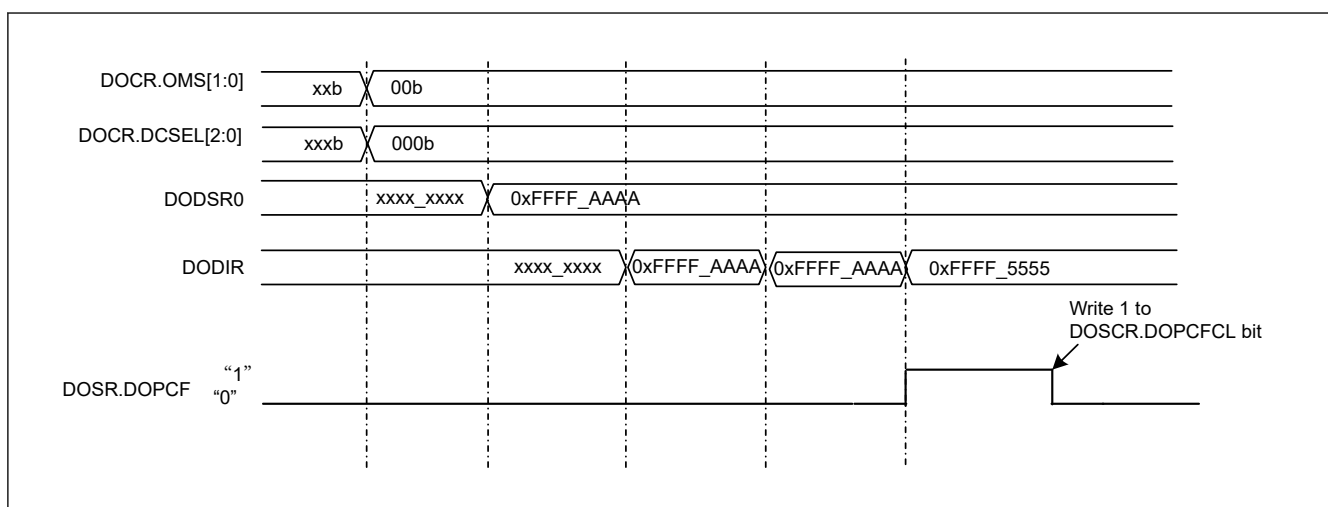


Figure 58.2 Example of Operation in Data Comparison Mode (Detection condition: Mismatch)

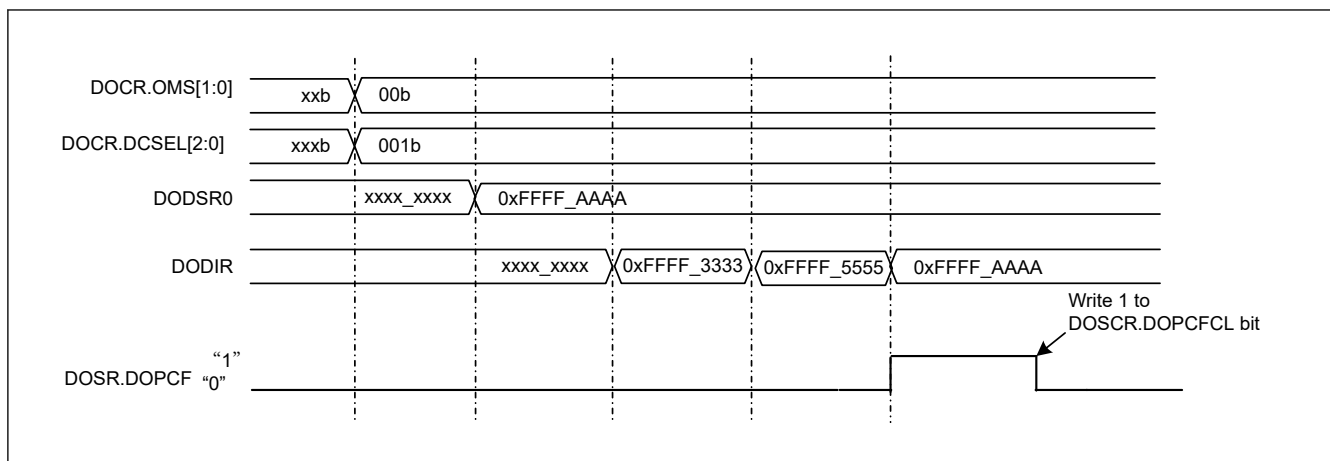


Figure 58.3 Example of Operation in Data Comparison Mode (Detection condition: Match)

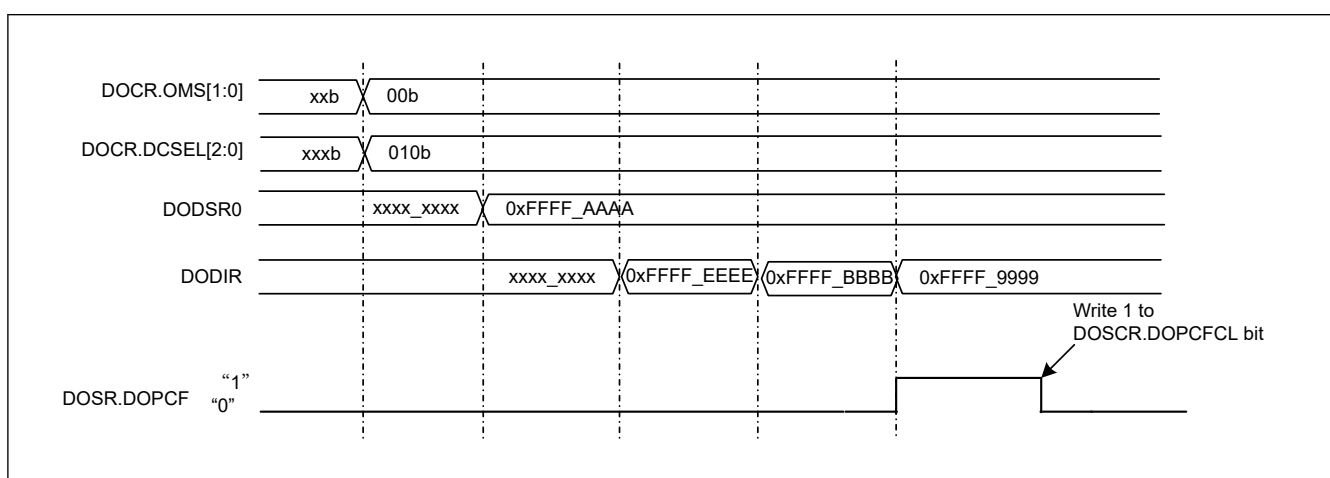


Figure 58.4 Example of Operation in Data Comparison Mode (Detection condition: Lower)

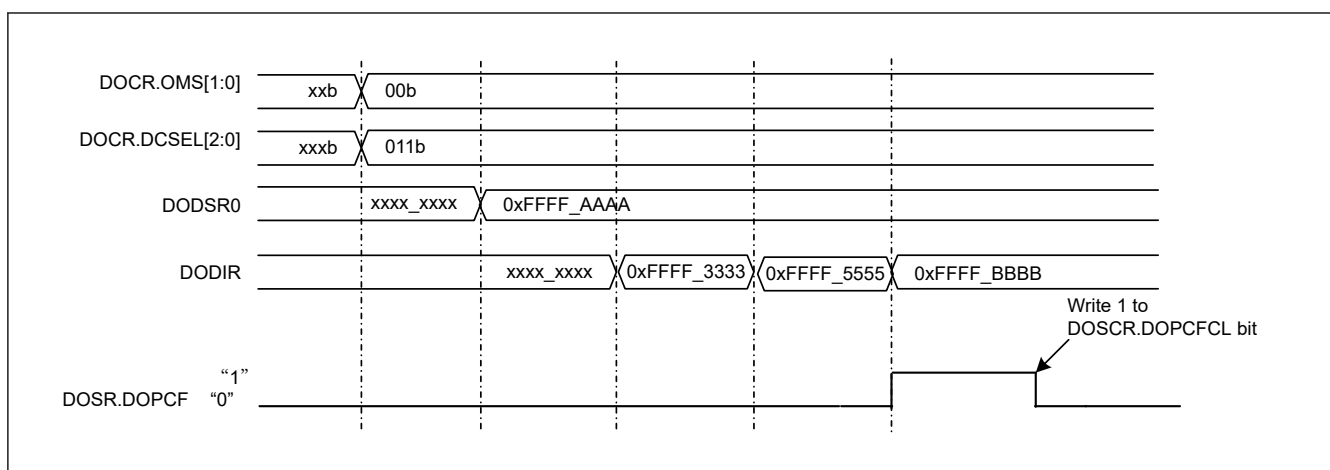


Figure 58.5 Example of Operation in Data Comparison Mode (Detection condition: Upper)

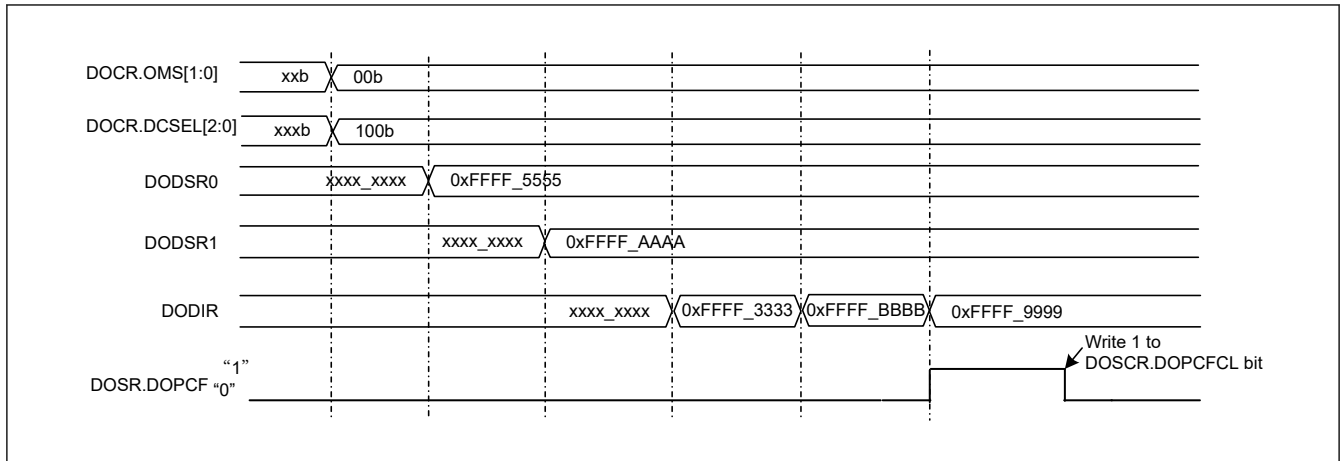


Figure 58.6 Example of Operation in Data Comparison Mode (Detection condition: Inside window)

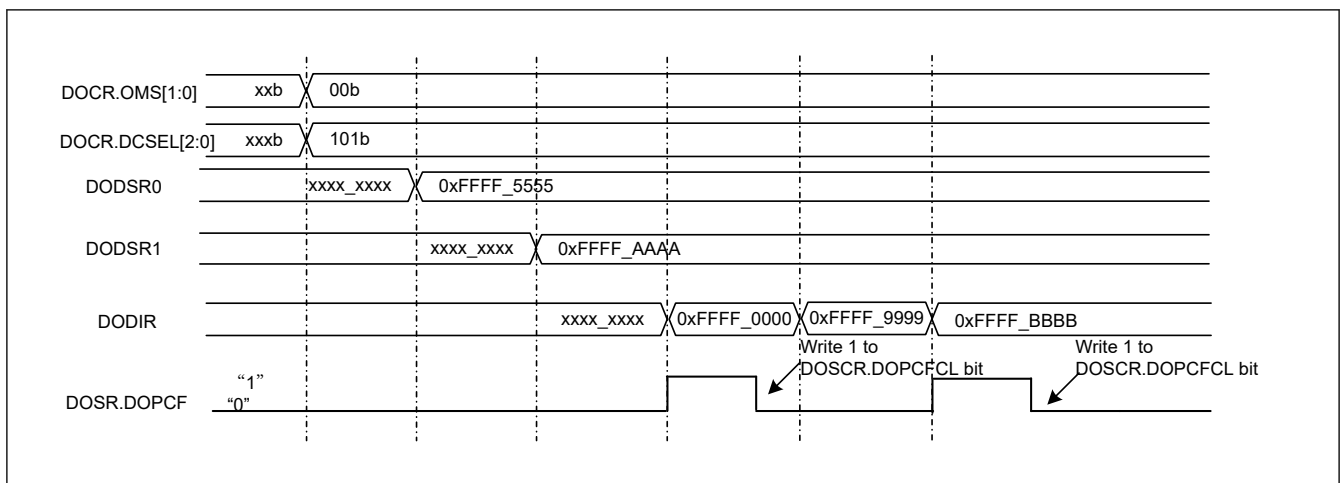


Figure 58.7 Example of Operation in Data Comparison Mode (Detection condition: Outside window)

58.3.2 Data Addition Mode

Figure 58.8 shows an example of the steps involved in data addition mode ^{*1} operation by the data operation circuit.

The following is an example of operation when the bit width of data operation is 32-bit.

1. Writing 01b to the DOCR.OMS[1:0] bits selects data addition mode.
2. 32-bit data is set in the DODSR0 register as the initial value.
3. 32-bit data to be added is written to DODIR. The result of the operation is stored in DODSR0.
4. Writing of 32-bit data continues until all data for addition have been written to DODIR.
5. If the result of an operation is greater than 0xFFFF_FFFF, the DOSR.DOPCF flag is set to 1 and an ELC event and a data operation circuit interrupt are generated.

Note 1. Addition is executed only by writing to the DODIR.

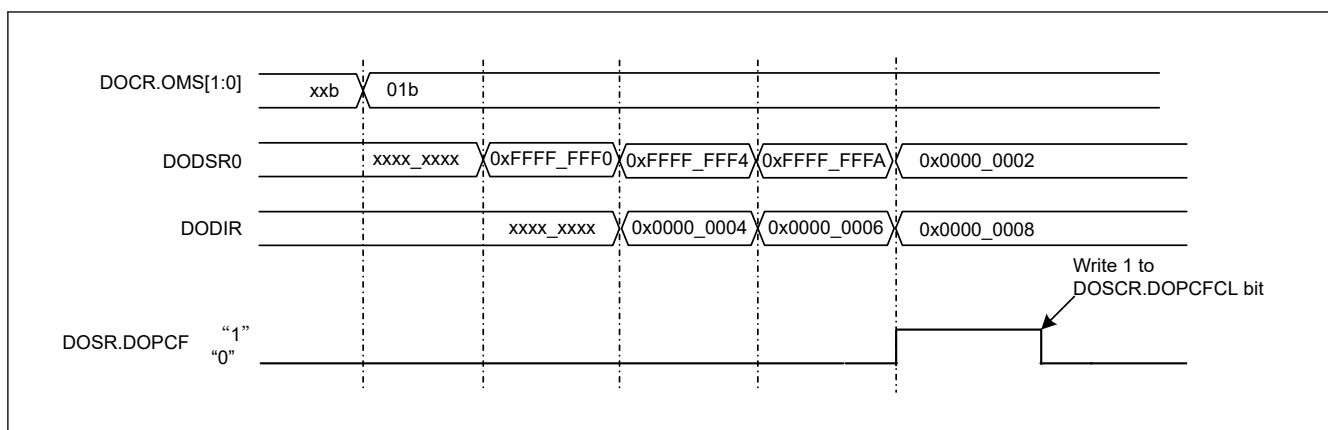


Figure 58.8 Example of Operation in Data Addition Mode

58.3.3 Data Subtraction Mode

Figure 58.9 shows an example of the steps involved in data subtraction mode ^{*1} operation by the data operation circuit.

The following is an example of operation when the bit width of data operation is 32-bit.

1. Writing 10b to the DOCR.OMS[1:0] bits selects data subtraction mode.
2. 32-bit data is set in the DODSR0 register as the initial value.
3. 32-bit data to be subtracted is written to DODIR. The result of the operation is stored in DODSR0.
4. Writing of 32-bit data continues until all data for subtraction have been written to DODIR.
5. If the result of an operation is less than 0x0000_0000, the DOSR.DOPCF flag is set to 1 and an ELC event and a data operation circuit interrupt are generated.

Note 1. Subtraction is executed only by writing to the DODIR.

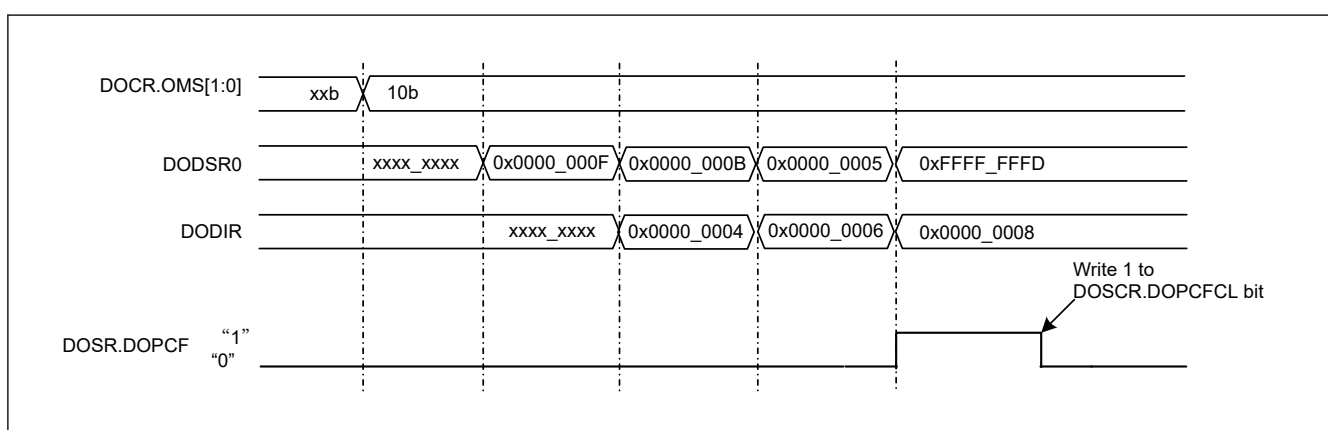


Figure 58.9 Example of Operation in Data Subtraction Mode

58.4 Interrupt Source

The data operation circuit generates the data operation circuit interrupt (DOC_DOPCI) as an interrupt request. When an interrupt source is generated, the data operation circuit flag corresponding to the interrupt is set to 1, and then interrupt request signal is generated. Table 58.2 describes the interrupt request.

Table 58.2 Interrupt request from DOC

Interrupt request	Status flag	Interrupt source
DOC interrupt	DOPCF	- The compared values match the detection condition. - The result of data addition is greater than 0xFFFF (DOCR.DOBW = 0) or 0xFFFF_FFFF (DOCR.DOBW = 1). - The result of data subtraction is less than 0x0000 (DOCR.DOBW = 0) or 0x0000_0000 (DOCR.DOBW = 1).

58.5 Event Link Output

The DOC outputs event signals for the event link controller (ELC) under the following conditions, and these can be used to initiate operations by other modules selected in advance.

- The compared values match the detection condition
- The data addition result is greater than 0xFFFF (DOCR.DOBW = 0) or 0xFFFF_FFFF (DOCR.DOBW = 1)
- The data subtraction result is less than 0x0000 (DOCR.DOBW = 0) or 0x0000_0000 (DOCR.DOBW = 1)

58.6 Interrupt Handling and Event Linking

The DOC has a bit to enable or disable interrupts. An interrupt request signal is output for the CPU when an interrupt source is generated while the corresponding enable bit is enabled.

In contrast, an event link output signal is sent to other modules as an event signal via the ELC when an interrupt source is generated, regardless of the setting of the corresponding interrupt enable bit.

58.7 Usage Notes

58.7.1 Settings for the Module-Stop State

The module Stop Control Register C (MSTPCRC) can enable or disable DOC operation. The DOC is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).